The rotor heads are run in a vacuum. The tube containing the homogenate is held at an angle to the axis. This technique is commonly used for collecting cells or producing subcellular fractions from a tissue homogenate. For example, homogenizing a rat liver at relatively low speed will pellet the larger dense nuclei and centrifugation at higher speed will sediment particles to further lower size (Fig.10.13)

ii. Density gradient centrifugation

This technique was developed by Scientist Brakke in the year 1960, where he proposed three ways through which density gradients may be used.

a. Rate zonal density gradient centrifugation:

This type of centrifugation involves, layering of sample solution on the top of the gradient, where the density continuously increases towards the bottom of the tube. It is used to measure the rate of sedimentation for separation of particles with different sedimentation co-efficient.

b. Isopycnic density gradient centrifugation:

Isopycnic (Equal) density gradient centrifugation depends on the buoyant density of the particle and not on the size and shape of the particle. This centrifugation is used for the separation of similar size particles which differ in their densities

c. Equilibrium iso-density centrifugation:

In equilibrium iso-density centrifugation, the sample is mixed with gradient medium to give a solution of uniform density. The gradient is self-forming during centrifugation. Salts of heavy metals like caesium, sucrose, and silica are used as the gradient material. Comparison of rate zonal density gradient centrifugation and isopycnic gradient centrifugation is represented in Fig.10.14

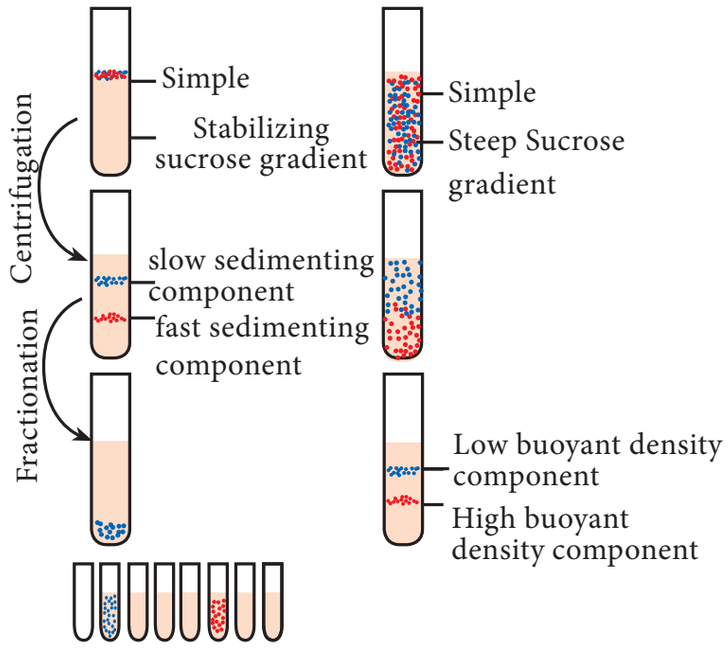

Figure 10.14 Equilibrium iso-density centrifugation

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